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Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems

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CSA1558 :CLOUD COMPUTING AND BIG DATA ANALYTICS FOR
ENGINEERS

Date: 04/06/2026

Technical Presentation



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Introduction

What is IoT?

- IoT refers to a network of interconnected physical devices.
- Devices collect and exchange data through the internet.
- Examples:
 - Smart watches
 - Smart home appliances
 - Sensors
 - Connected vehicles

What is Cloud Computing?

Provides storage, processing, and computing resources over the internet. Enables scalable and on-demand services.



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Core Technical Explanation

Integration of IoT and Cloud Computing

IoT devices generate huge amounts of data.

Cloud computing:

Stores data.

Processes information.

Performs analytics.

Enables remote access.

Supports machine learning and AI applications.

Working Process

Sensors → Internet → Cloud Platform → Data Processing → User Application



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IoT-Cloud Architecture

Components of IoT-Cloud Architecture

1. IoT Devices

Sensors

Cameras

Wearables

2. Communication Network

Wi-Fi

Bluetooth

5G

ZigBee

3. Cloud Platform

Data storage

Computing resources

Analytics

4. Application Layer

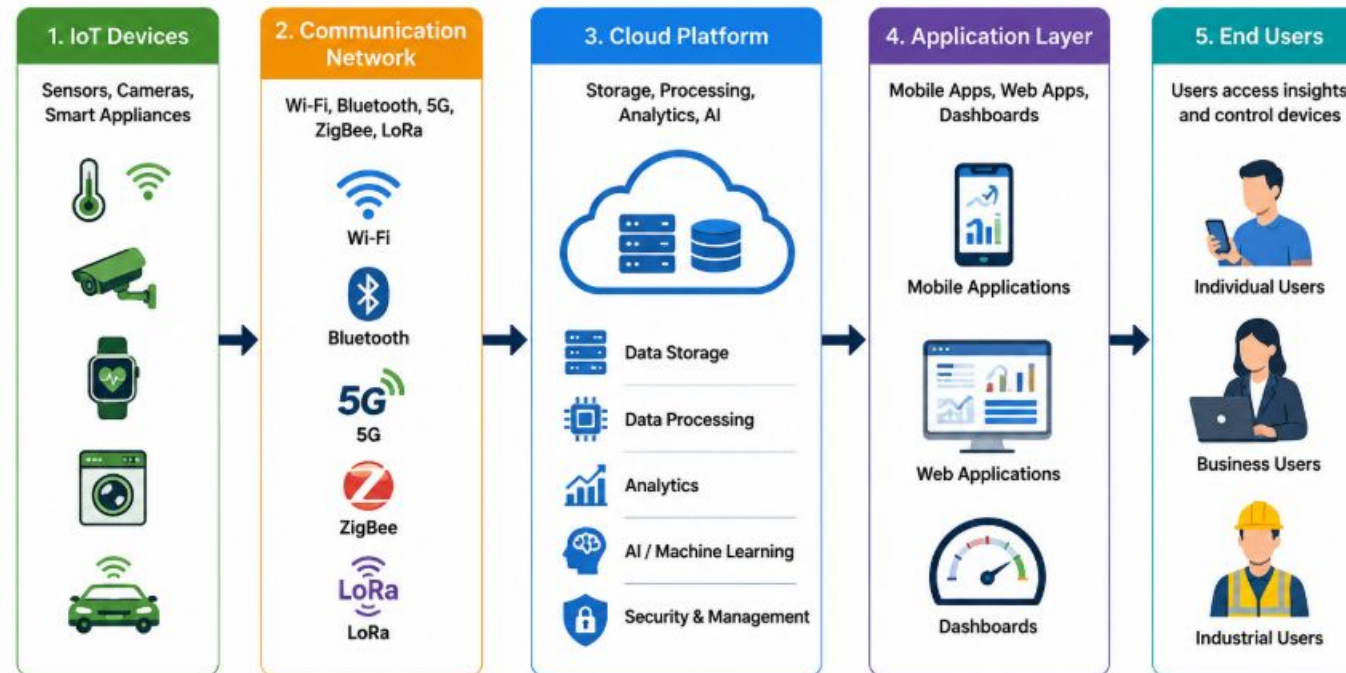
Mobile applications

Dashboards

Web interfaces

Architecture Diagram

IoT-Cloud Architecture





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Benefits of IoT and Cloud Integration

Scalability

Easily handles millions of connected devices.

Real-Time Monitoring

Provides instant information and alerts.

Cost Efficiency

Reduces infrastructure costs.

Data Analytics

Supports AI and predictive analysis.

Remote Access

Users can control devices from anywhere.



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Real-World Applications

Smart Homes

Smart lights

Smart thermostats

Security cameras

Healthcare

Remote patient monitoring

Wearable devices

Smart Cities

Traffic management

Smart parking systems

Industrial IoT (IIoT)

Predictive maintenance

Factory automation

Agriculture

Soil moisture monitoring

Smart irrigation systems



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Example Use Case

Smart Healthcare System Process

Patient Wearable Sensors



Internet Connectivity



Cloud Storage



AI-Based Analysis



Doctor Dashboard



Real-Time Monitoring and Alerts

Advantages

Continuous health monitoring.

Early disease detection.

Reduced hospital visits.

Improved patient care.



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Conclusion and Key Takeaways

Conclusion

IoT and Cloud Computing together create intelligent connected ecosystems.

Cloud platforms provide storage, processing, and analytics for IoT data.

They enable automation and smart decision-making across various industries.

Key Takeaways

- ✓ IoT devices collect data.
- ✓ Cloud computing stores and processes data.
- ✓ Integration improves scalability and efficiency.
- ✓ Applications include healthcare, smart homes, industries, and agriculture.
- ✓ Security and privacy remain major challenges.



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Limitations and Future Scope



Thank You!!!